

# Amirhossein Rezaei

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## EDUCATION

### University of Padova

*Physics of Data (M.Sc.)*

Padova, Italy

*Oct 2025 – Present*

### Shahid Beheshti University

*B.Sc. in Physics; GPA: 3.54/4.00*

Tehran, Iran

*Sep 2019 – Feb 2024*

### Shahid Madani High School, NODET

*National Organization for Development of Exceptional Talents*

Tabriz, Iran

*Sep 2013 – Jun 2017*

## RESEARCH EXPERIENCE

### Research Papers

- Continuous Approximation of the Ising Hamiltonian: Exact Ground States and Applications to Fidelity Assessment in Ising Machines. *Physical Review E*, 112(2), 024102. [↗](#)  
*An intuitive and very accessible video explanation of the paper's key concepts and results is available on [YouTube](#): [Watch here](#)*
- The generation of baryon asymmetry and hypermagnetic field by the chiral vortical effect in the presence of sphalerons. *Nuclear Physics B* (2025): 116895. [↗](#)
- Effects of Nonhelical Component of Hypermagnetic Field on the Evolution of the Matter-Antimatter Asymmetry, Vorticity, and Hypermagnetic Field. *Physical Review D* 104 (5). [↗](#)
- A Deep Learning Framework for Collapse Prediction of Structures Using 1D-CNN for Earthquake Time Series Classification with joint Bayesian hyperparameter optimization and Neural Architecture Search. (*Under review*. Pre-print available [here](#).)

### Teaching Assistant

- Physics I, Assistant to Dr. Sadeghi

## RELEVANT COURSES AND GRADES

|                                            |          |
|--------------------------------------------|----------|
| Quantum Information and Computations       | 16.8/20  |
| Quantum Mechanics III                      | 18.51/20 |
| Introduction to Artificial Intelligence    | 17.5/20  |
| Biophysics                                 | 20/20    |
| Complex Systems                            | 19/20    |
| Mathematical Physics II                    | 19.5/20  |
| Physics Project                            | 20/20    |
| Foundations of Numerical Simulations       | 20/20    |
| Nuclear and Elementary Particle Physics    | 17.51/20 |
| Thermodynamics and Statistical Mechanics I | 17.3/20  |
| Analytical Mechanics I                     | 17.75/20 |
| General Topology                           | 14.5/20  |

For the full grades, see [here](#).

## SKILLS

**Programming:** Python, C, C++, Fortran, MATLAB, Mathematica, Maple

**Tools and Technologies:** PyTorch, TensorFlow, Keras, Scikit, SciPy, Pandas, MNE, Matplotlib, NumPy, Git, L<sup>A</sup>T<sub>E</sub>X

**Languages:**

- Persian (Native)
- English (C1 – IELTS Overall Band 7.5; Writing: 7.5, Listening: 7.5, Reading: 8, Speaking: 7)
- Azerbaijani
- Turkish

## PROJECTS

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### Gargantua: Cinematic Black Hole Renderer | [GitHub](#)

- Developed a real-time, cinematic black hole renderer in Python, leveraging the Taichi language to JIT-compile high-performance CUDA kernels for the GPU.
- Engineered a physics-based backend featuring a general relativistic ray tracer (using a Dormand-Prince 5(4) adaptive integrator) and a custom 3D Eulerian fluid simulation on a cylindrical coordinate grid for the accretion disk.
- Integrated advanced volumetric rendering, a full cinematic post-processing suite (ACES tone mapping, bloom, anamorphic flares), and an intelligent frame caching system to optimize workflow and visual fidelity.

### SoME4 Video Project | [GitHub](#)

- An animated explainer video for the 2025 Summer of Math Exposition (SoME4), built with [Manim](#) and [PsiCreature](#), showcasing research in Ising machine fidelity through storytelling and custom character animation.
- The video explains the paper “*Continuous Approximation of the Ising Hamiltonian: Exact Ground States and Applications to Fidelity Assessment in Ising Machines*”.
- The YouTube video is available [here](#)

### PsiCreature | [GitHub](#)

- PsiCreature: A modular character animation framework for Manim, featuring expressive SVG-based body templates, squinting eyes, bending sclera, and dynamic emotional states.

### Divar Scraper with Multithreading | [GitHub](#)

- Divar Scraper with Multithreading: This project allows you to scrape data from the Divar website, specifically for real estate listings in different categories and cities. It utilizes the Divar API to retrieve and store information about real estate listings in an Excel file. This module provides a command-line tool for scraping posts from a website.

### TGJU-API | [GitHub](#)

- TGJU-API: A Python web scraper for retrieving currency and gold prices from the tgju.org website.

### Complex-Systems | [GitHub](#)

- A repository of Complex Systems ideas and codes.

### Optimizing the Ising model in Python | [Medium](#)

- This project involves simulating a 2D ISING model using various techniques such as utilizing the fast Numba JIT Compiler, implementing parallel processing with Numba's prange, performing branch-less computation, utilizing a lookup table for the exponential function, and optimizing further with a custom random number generator.

## RELATED ACTIVITIES

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### *Physics Student Association of Shahid Beheshti University (2022-2023)*

- Elected Member, Main Council of Physics Student Association, Shahid Beheshti University
- Volunteered and campaigned for the position, receiving support from students through votes.
- Coordinated events and activities for students interested in physics and related fields.
- Organized weekly seminars and workshops featuring guest speakers from academia and industry.
- Collaborated with other student organizations to promote science education and outreach in the community.

### *Representative of the Faculty of Physics and Adviser to the University President (2023-2024)*

- Selected Member, Representative of the Faculty of Physics, acting as a voice for the student body.
- Actively involved in advocating for student demands and interests within the faculty.
- Played a key role in enhancing the welfare and academic environment of the Physics department.
- Organized and coordinated events, seminars, and workshops for students and faculty.